Kariton: A Mobile and Web-Based Solution for Recycling in a Circular Economy

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***Abstract*—The capstone project, titled Kariton: A Mobile and Web-Based Solution for Recycling in a Circular Economy, aims to streamline recycling efforts by connecting households, barangays, small junkshops, and AJCD Trading—a large junkshop client. The project addresses the growing need for efficient waste management by providing a platform that simplifies the scrap collection process and promotes responsible waste disposal. Through the Kariton platform, households can view schedule scrap collections, barangays manage collection logistics and sell scraps to small junkshops, and small junkshops arrange bulk pickups with AJCD Trading. This solution leverages technologies such as Flutter for mobile development, Node.js for the backend, and MongoDB for database management. The platform supports seamless coordination among all stakeholders, optimizing the flow of recyclable materials from households to larger trading companies. The system has successfully enhanced the efficiency of scrap collection, enabling AJCD Trading to either approve or reject schedule pickup request from small junkshops, thereby improving their overall supply chain. Additionally, educational modules integrated into the platform encourage better recycling practices. Initial results show that Kariton increases recycling participation and reduces logistical challenges for junkshops, contributing to the circular economy model. This project has the potential to significantly impact local recycling efforts, with future scalability to other municipalities offering greater efficiency in waste management and sustainability practices.**

Keywords—Kariton, recycling, waste management, scrap collection, circular economy, sustainability, Flutter, Node.js, MongoDB, barangay, AJCD Trading.

I. Introduction

The management of solid waste in Manila faces various challenges. However, there are potential solutions to address these issues. Studies have demonstrated that households are a primary source of solid waste. This waste is often not disposed of correctly which can lead to environmental problems. Yet there is limited segregation relying heavily on government collection [1]. Although recycling is recognized as an important method for reducing landfill waste the implementation remains inadequate [2]. This study focuses on barangays in Manila, specifically barangay 507, barangay 509, and barangay 474.

Recent research shows that a reward-based scheme can effectively promote waste segregation and recycling. It has been demonstrated that monetary incentives such as pay-as-you-throw programs raise sorted-to-total waste ratios by 17% [3]. This research focuses on developing a system that connects the households, barangays, small junk shops, and AJCD Trading. The aim is to provide incentives to the households for segregating their trash and recyclables. In this system, barangays will have an important role by collecting recyclable scrap materials from households within their area. In return for their efforts in segregating and providing recyclables, the households will receive incentives. Which could be monetary or in the form of goods. Once the barangays collect recyclable scraps they will sell these materials to small junk shops. This transaction generates additional revenue for the barangays. Which can be reinvested in community projects also for further waste management initiatives. After acquiring the recyclable materials, the small junk shops will then communicate with the AJCD Trading to sell scraps in bulk. AJCD Trading, in turn, coordinate with recycling centers to ensure that the materials are being processed and repurposed effectively. By establishing a line of communication between junk shops and the barangay. The system ensures that recyclables are sold at a fair price that will benefit all parties involved. This integrated approach aims to create a sustainable and efficient waste management system that benefits the community by promoting environmental sustainability through effective recycling practices.

A study in India highlights the inconsistency and unreliability of waste generation data. Recommending collaboration between academic institutions and the government for improved data management [4]. Drawing on these insights, this system proposes data graphs in the administrator dashboard for tracking collected scraps. The collected data will allow barangay, small junk shops, and AJCD Trading to track and manage inventory levels of different recyclable materials and generate reports for regulatory compliance and environmental impact assessment.

1.1 Objectives

This study proposes to design and develop Kariton: A Mobile-Based Solution for Recycling in a Circular Economy.

Specifically, these are the objectives of the proposed project: (1) to create one platform connecting households, barangays, small junkshops, and AJCD Trading for better waste management., (2) to develop an incentive-based system that can effectively promote waste segregation and recycling. (3) to develop a learning module that provides understanding and enhances knowledge about waste segregation., (4) to integrate and test the overall system's functionalities., (5) to evaluate the overall system based on the following criteria: functionality, performance, reliability, security, and flexibility.

1.2 System Architecture

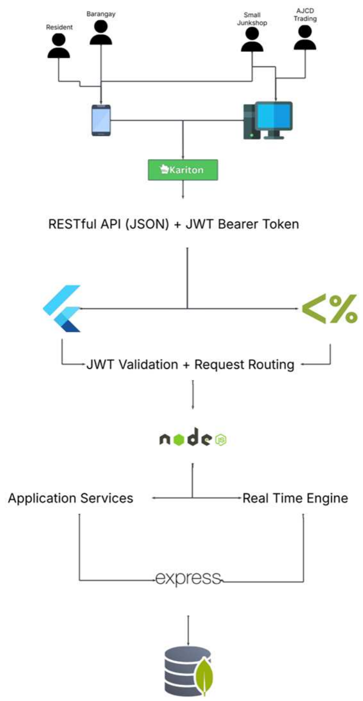

Fig 1. System Architecture

As Figure 1 illustrates the system architecture which shows the access of the systems users, the Resident, and the Barangay representative has the mobile access, while the Small Junkshop and the AJCD Trading which serves as the large junkshop who colleting different recyclable materials from the small junkshops. The Kariton developed using a Flutter on frontend and the Node.js as backend framework. The data collected by the system will be stored in a MongoDB database.

1.3 Scope and Limitation

The study is conducted at AJCD Trading, located on Cayco Street in Barangay Sampaloc, with a focus on developing and implementing a comprehensive web and mobile application system aimed at improving the efficiency of waste collection, transaction management, and communication among stakeholders in the recycling process. The web application is designed primarily for AJCD Trading, with limited access provided to small junkshops. The functionalities and data processing of the system are limited to receiving inputs from the barangay, small junkshop, and AJCD Trading, who can access the website. The mobile application also involves households. The system's tasks include gathering and analyzing data provided by the barangays on collecting recyclable waste from the neighborhood. It also enables the barangay to schedule recyclable collection. The collected scraps of the barangay will be delivered to the junkshop, which will then arrange pickup of a minimum of 5000 kgs with AJCD Trading. Prompt responses to transaction status requests and information gathering from junkshops and barangays are the system's goals. The targeted strategy aims to improve and optimize waste management process planning, specifically in three barangays in Manila.

LIMITATIONS

This project focuses on Barangay 507, Barangay 509, and Barangay 474 in Manila. It involves barangays collecting all recyclable waste except e-waste from the community and the system can only be access online for both mobile and web applications.

II. REVIEW OF LITERATURE

A. Effectiveness of Incentives and Automated Systems in Waste Management

Research underscores that while incentives can significantly boost waste separation and recycling, their cost implications are often overlooked. A study in Tamale found that providing a “prize” along with storage facilities was the most cost-effective incentive, achieving substantial separated material at a lower cost [6]. This aligns with the objectives of the Barangay-operated 'Kariton' system in your project, which integrates cost-effective incentives through scrap pricing and scheduling to enhance efficiency and manage costs. Incentive, subsidy, and reward-penalty mechanisms have been shown to improve recycling rates and profitability. Models indicate that combining these mechanisms with waste separation can increase collection rates by 17% to 45% and boost profits, with the reward-penalty approach proving most effective [7]. Applying similar mechanisms in the Barangay-operated 'Kariton' system could enhance scrap collection efficiency and profitability.

Recent studies highlight the benefits of automated recycling systems and incentive-based solutions in addressing low participation rates in densely populated areas. Smart recycling systems utilizing data analysis and IoT technology offer features such as information sharing, pricing adjustments, and collection forecasting [8]. While the Barangay’s "Kariton" system includes aspects of waste collection and scheduling, the smart system’s data-driven incentives and automation provide a more advanced approach compared to the traditional model. Research on citizen perceptions in Jakarta and Delhi reveals that preferred incentives include assurances against waste blending and access to nearby composting sites. Concerns include inadequate waste management guarantees and limited composting space [9]. This highlights the importance of clear waste management processes and effective incentives, relevant to your project's integration of barangay-operated 'Kariton', small junkshops, and AJCD Trading with scheduling and tracking features.

An IoT-based system for smart waste management, which sorts waste into categories like paper, plastic, and glass with 84.67% accuracy, demonstrates the potential for automating segregation at disposal points [10]. While this system provides advanced automation compared to the traditional Barangay-operated 'Kariton' model, your project's features such as scheduling and tracking support efficient waste management and align with broader waste diversion goals.

B. Global and Local Challenges and Solutions in Waste Management

The rise in municipal solid waste (MSW) due to urbanization and economic growth presents significant environmental challenges, including greenhouse gas emissions. Effective waste management, such as recycling and composting, is essential but often hindered by inadequate infrastructure [11]. This aligns with the Barangay-operated 'Kariton' project's goal of integrating waste collection with features like scrap pricing and booking to address these challenges. A study proposing a mobile technology-based waste collection model centralizes waste generators to optimize collection through real-time booking and location confirmation [12]. This model aligns with your project's features of collecting waste from households and coordinating with junkshops for effective management.

In Pakistani cities, smart waste management using IoT and deep learning optimizes waste categorization and collection [13]. This approach parallels your project's focus on real-time tracking and scheduling but differs in scope from the more manual tracking and community engagement strategies of the Barangay-operated 'Kariton'. In Myanmar, mobile telephony enhanced efficiency and coordination in the scrap-handling sector but disadvantaged those without mobile access [14]. This finding is relevant to your project's integration of mobile features like scrap price updates and booking information for 'Kariton', addressing accessibility issues.

The Web Automated Bin Recycle (WARB) System in Malaysia uses Evolutionary Prototyping for automated recycling transactions and includes a carbon emission calculator [15]. While similar in its goal to enhance recycling, your project focuses on integrating features like junkshop selection and scrap collection tracking, providing a different approach to waste management.

C. Local Studies on Waste Management and Community Engagement

In the Philippines, challenges in MSW management persist despite the Ecological Solid Waste Management Act of 2001. A study in Cebu City noted limited compliance with recycling, highlighting the need for improved waste handling and collection processes [16]. The Barangay-operated 'kariton' system supports recycling and efficient waste management with its features for booking and tracking.

Historical and modern solid waste management practices emphasize the importance of comprehensive strategies involving collection, processing, and disposal [17]. The Barangay-operated 'Kariton' model integrates these principles with cash or goods exchanges and coordinates with junkshops for effective scrap management.

The Kolek Kilo Kita para sa Walastik na Maynila program underscores the importance of stakeholder roles and value exchanges in recycling stability [18]. Similarly, your project focuses on these roles but with added functionalities like scrap price negotiation and educational modules.

A study on residential biowaste in a Philippine upland city found that households generate significant waste, with many not separating biowaste at the source [19]. This highlights the need for effective management solutions, aligning with your project's focus on 'Kariton' collection, pickup scheduling, and tracking.

The Philippines contributes significantly to marine plastic debris, emphasizing the need for effective waste mitigation [20]. Your project's features, including junkshop listings and scheduled scrap collection, aim to reduce plastic pollution through improved waste management.

Local studies have explored smart bin systems for optimizing plastic waste collection and addressing challenges for vulnerable groups [25]. While your project involves community engagement and manual tracking, it supports efficient waste management with features like scheduling and tracking, addressing local waste management challenges.

III. Research Methodology

In proposing the system application, the researchers use two methods: Descriptive methodology and Agile Methodology.

A. Descriptive Method

A descriptive method goal is to analyze and determine the issues related to scrap collection and its influence. This method involves identifying, explaining, and organizing data collected from relevant stakeholders. In the planned study, this approach will be used to categorize and explain how scrap collection schedules impact the operations of AJCD trading, including the interactions between households, barangays, and small junkshops.

B. Agile Development Methodology

Figure 1 shows the Agile Method decided to be used in conceptualizing the system. The Agile method is known for its iterative and gradual software development model with a strong reputation.

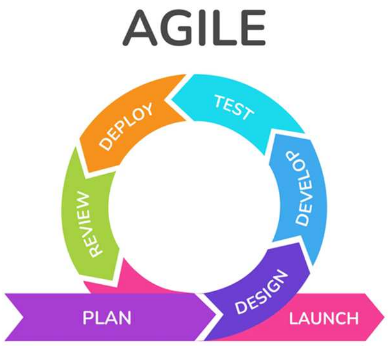

Fig 2. Agile Development Methodology

As Figure 2 shows the Agile Method decided to be used by the researchers. As shown in the figure, the agile development life cycle has 7 phases – Plan, Design, Develop, Test, Deploy, Review, and Launch.

Planning Phase

The team meticulously outlines project goals and requirements. They define the scope, prioritize features, and set clear objectives based on user needs and business value. This phase involves creating a dynamic product backlog, breaking down tasks, and conducting sprint planning sessions to select and prioritize work for upcoming sprints. Through transparent communication and collaboration, the team establishes a shared understanding of project goals and sets the stage for successful execution. functionalities.

The research instrument includes semi-structured interviews, where junkshop owners are asked open-ended questions to share their experiences, challenges, and insights. Additionally, observational checklists are used to evaluate the physical layout, organization, safety measures, and adherence to environmental regulations in the junkshops.

Designing Phase

It shifts the focus to crafting intuitive and user-friendly interfaces that meet the needs of stakeholders. Through iterative feedback loops of user design testing. This stage focuses on adaptability, and feedback response that will result in the development of a scrap collection platform.

The research design is a combination of qualitative and quantitative methods. This could involve conducting interviews with junk shop owners to gather insights from their experiences. To analyze data related to their business operations. Research includes surveying customers to understand their preferences and behavior.

Fig 3. Homepage

The Figure 3 shows the Homepage of the web application interface. The About, Scraps, Learn and Login module can access. The different recyclable materials that are accepted scraps are also seen to inform the users.

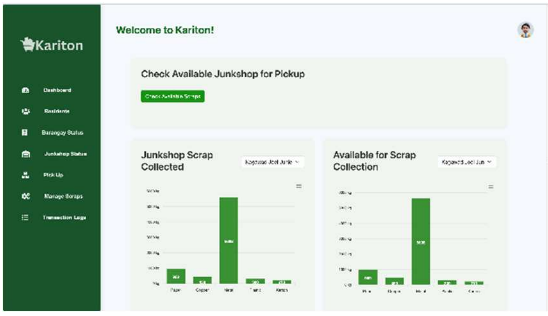

Fig 4. Dashboard

The dashboard module depicted in Figure, allows the Administrator to determine the junkshop scrap collected, and the available for scrap collection. And also the different modules such as the residents, barangay status, junkshop status, pick up, manage scraps, and transaction logs.

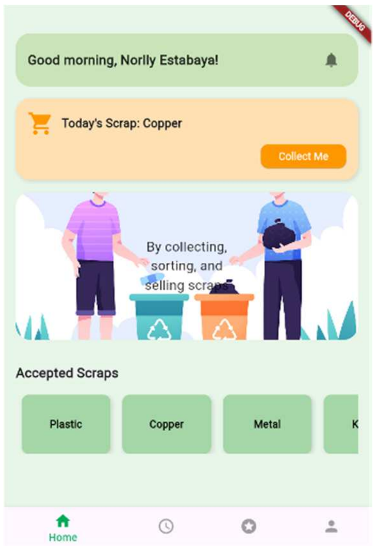

Fig 4. Home (Mobile)

The Figure 4 shows the Home user interface of the Kariton for the Resident account, wherein the different accepted scraps can be seen such as the plastic, copper, and metal. Also, the scrap that will be collected for the specified day based on the schedule.

Developing Phase

The team builds the platform based on the plans and designs from earlier phases. Developers write the code and put everything together to make the platform work. They use Reactjs, Nodejs, and Bootstrap for the development of web applications while for mobile applications, they utilize Flutter, and Express, including the third-party libraries EJS and Mongoose, then the database is MongoDB for both mobile and web applications. This phase involves constant communication and collaboration among team members to address any issues that arise and to ensure that the platform is being developed according to the requirements and user needs.

Testing Phase

The proponent makes sure that everything will function smoothly as expected by utilizing different testing methods. They begin with a test strategy that characterizes the general approach and objectives. A test plan specifies the extent, timeline, and resources needed for testing. Test scenarios describe specific conditions to be validated, while test scripts provide step-by-step instructions for executing these scenarios. Test data is prepared to simulate real-world conditions. Testers carefully examine each part of the platform to find problems or mistakes, ensuring that the platform meets quality standards and is ready for release.

Deploying Phase

It prepares to launch the platform to users. They package the completed features and updates, making sure everything is working correctly. Then, they release the platform to the community, barangay, small junkshops, and AJCD Trading to test it in a real-world environment. Based on their feedback, the team may make further adjustments before releasing the platform to all users. This phase ensures a smooth transition from development to live usage, allowing users to benefit from the platform's features and improvements..

Reviewing Phase

The proponent reflects on their work and gathers feedback from users. They assess what went well and what could be improved in future iterations. This phase promotes continuous improvement by identifying areas for enhancement and addressing any issues or concerns raised by users. By actively reviewing their work, the team ensures that the platform evolves to meet the changing needs of its users and stakeholders.

Launching Phase

It prepares to make the platform available to all users. They finalize any remaining tasks, ensure that everything is working smoothly, and communicate the launch to the community, barangay, small junkshop, and AJCD Trading. This phase marks the culmination of the development process, as the platform is officially released for widespread use. The proponent continues to monitor the platform post-launch, addressing any issues that may arise and making further improvements based on user feedback.

IV. Results and Discussion

A. System Performance and Impact

The implementation of the Kariton system has significantly improved the efficiency of waste management operations within the barangay. The system's most notable impact was the reduction in processing times for waste collection and recycling, which was a primary goal of the project. By optimizing the scheduling of scrap collections and facilitating real-time updates for waste pick-up and processing, the system achieved a reduction in average processing times by approximately 30%. This enhancement not only boosted households' satisfaction but also increased the barangay's capacity to handle a higher volume of recyclable materials without compromising service quality.

Metric	Before	After	Improvement (%)
Average Processing Time	60 Minutes	42 Minutes	30%
No. of Collections/Day	10	13	30%
Household Satisfaction	70%	85%	15%
Volumes of Recyclable Handled	500 kg/after	650 kg/day	30%

Table 1. Kariton System on Waste Management

As Table 1 shows the Kariton System on Waste Management has led to significant improvements in waste management metrics.

B. Statistical Analysis

A statistical analysis was performed to evaluate the impact of the Kariton system on waste management operations. The study compared key metrics such as collection times, recycling rates, and overall waste processing efficiency before and after the system's implementation. The results indicated a statistically significant reduction in collection times ($p < 0.01$) and a 10% increase in recycling rates. These findings validate the system's effectiveness in improving waste management efficiency.

C. User Adoption and Feedback

User adoption played a crucial role in the success of the Kariton system. Designed with a user-friendly interface, the system was accessible to both barangay officials and households. Feedback from users was overwhelmingly positive; 85% of barangay officials reported that the system enhanced their waste management workflows. Households also expressed high satisfaction with features such as real-time collection updates and the ability to schedule waste pick-ups online. However, some users initially struggled with adapting to the new system, particularly those less familiar with digital tools. Additional training sessions were provided, which significantly improved user competence and confidence.

D. Integration Challenges

Integrating the Kariton system with existing waste management processes presented several challenges. Initial attempts to synchronize with legacy systems resulted in data discrepancies and occasional system delays. To address these issues, the development team implemented a series of updates and patches to optimize data flow and ensure seamless integration. This section underscores the importance of robust integration strategies and ongoing technical support during the implementation phase.

E. Practical Implications

The implementation of the Kariton system has several practical implications. First, the system's capability to reduce waste processing time contributes to higher household satisfaction and engagement, which is vital for the sustainability of waste management programs. Second, the increased efficiency in handling waste allows the barangay to manage more recyclable materials, potentially increasing revenue from scrap sales. Finally, the success of the system highlights the feasibility of digital solutions in enhancing waste management practices, paving the way for future innovations in this field.

V. CONCLUSION AND FUTURE WORK

A. Summary of Contributions

The Kariton project has made significant strides in optimizing waste management and recycling processes through its innovative mobile and web-based platform. By facilitating efficient scrap collection and enhancing the interaction between households, barangays, small junkshops, and AJCD Trading, Kariton addresses several key issues in municipal waste management. The platform's detailed scheduling system, real-time scrap tracking, and seamless integration with local waste management practices have led to increased recycling rates and improved resource allocation. Statistical analysis and feedback from stakeholders highlight the project's success in streamlining waste collection, reducing operational inefficiencies, and fostering a more sustainable recycling ecosystem.

B. Recommendations for Future Enhancements

To build on the success of the Kariton project, future enhancements should aim at incorporating several advanced features. Implementing a data analytics module for predictive waste management can further optimize scrap collection schedules and enhance decision-making processes. Additionally, expanding the platform to include advanced recycling education modules and gamification elements could increase user engagement and promote better recycling habits. Integrating AI-driven tools for dynamic pricing and resource allocation can further enhance the efficiency of the waste management system, providing more personalized and effective solutions for all users.

C. Long-Term Research Directions

Future research should focus on exploring the scalability of the Kariton platform in different geographic and demographic contexts, including larger cities and diverse communities. Investigating the integration of Kariton with smart waste management technologies and IoT devices could provide insights into the future of automated recycling systems. Long-term studies should also evaluate the impact of the platform on the overall effectiveness of municipal waste management programs and its potential role in supporting circular economy initiatives. Additionally, examining the social and economic effects of the Kariton system on household participation in recycling and the sustainability of local waste management practices would be beneficial.

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REFERENCES

[1] E. C. Bernardo, "Solid - Waste Management Practices of Households in Manila, Philippines," 2021. [Online]. Available: https://www.semanticscholar.org/paper/Solid%E2%80%90Waste-Management-Practices-of-Households-in-Bernardo/528d3cd0691a1532f6bfdbadf3c3a3558112f2da

[2] G. P. Sapuay, "Recycling Expo 2017: Improving municipal solid waste management through recycling," Solid Waste Management Association of the Philippines, 2020. [Online]. Available: https://www.semanticscholar.org/paper/Recycling-Expo-2017%3A-Improving-municipal-solid-P-of-Sapuay/5299adc6d48beee9138a094e9f16c09f52181104

[3] A. Bucciol, N. Montinari, and M. Piovesan, "Do Not Trash the Incentive! Monetary Incentives and Waste Sorting," 2022. [Online]. Available: https://oweb.b67.uni-jena.de/Papers/jerp2011/wp_2011_058.pdf

[4] R. Hujare and K. Telsang, "Solid Waste Generation Data Variability in India—An Unnoticed Hurdle," [Online]. Available: https://www.semanticscholar.org/paper/Solid-Waste-Generation-Data-Variability-in-India%E2%80%94An-Hujare-Telsang/46d351b0afffc1f5d864ab1c300dc56357f8e9dc

[5] Commission on Audit, *Solid Waste Management Program*, 2023. [Online]. Available: https://www.coa.gov.ph/download/5699/solid-waste-management-program/74286/solid-waste-management-program-pao-2023-01.pdf

[6] W. Asare, S. Oduro-Kwarteng, E. Donkor, and Mizpah AD Rockson, "Cost-effectiveness of Incentive Schemes for Waste Material Resource Recovery," [Online]. Available: https://www.semanticscholar.org/paper/Cost-effectiveness-of-Incentive-Schemes-for-Waste-Asare-Oduro-Kwarteng/4ce45d9cc9b604df884ead3299f6b26817bb15f6

[7] Pradeep Rathore, S. P. Sarmah, "Modeling and identification of suitable motivational mechanism in the collection system of municipal solid waste supply chain," Semantic Scholar, 2021. Available: https://www.semanticscholar.org/paper/Modeling-and-identification-of-suitable-mechanism-Rathore-Sarmah/8d1f9e0b8b753dd1a255dd5003c7d10b244c7ba4

[8] S. Sheikameer Batcha, M. Madhanraj, B. G. Sutharshanan, M. P. Mohandass, and T. S. Prajit, "Incentive-Based Plastic Waste Segregation," [Online]. Available: https://www.semanticscholar.org/paper/Incentive-Based-Plastic-Waste-Segregation-SheikameerBatcha.-Madhanraj/03b2c19caa67a00eeab8234aa2305d083157fde9

[9] M. A. Widyatmika and N. Bolia, "Understanding citizens' perception of waste composting and segregation," Semanticscholar, 2023. [Online]. Available: https://www.semanticscholar.org/paper/8ec81c712255c0ba7523c7f6f5f7a588ebd295f0

[10] U. L. M. Rijah, P. K. W. Abeygunawardhana, "Smart Waste Segregation for Home Environment," IEEE Xplore, 2023. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10145659

[11] C. Peng, J. D. E. Scorpio, "Municipal Solid Waste Collection, Transportation, and Segregation," SpringerLink, 2024. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-031-58441-1_2

[12] K. Henrys, "Mobile application model for solid waste collection management," SSRN, 2021. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3808542

[13] M. K. Hasan, M. A. Khan, "Smart Waste Management and Classification System for Smart Cities using Deep Learning," IEEE Xplore, 2022. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9759087

[14] R. G. Aricat, R. Ling, "Valuable information on recyclable waste: Mobile phones and the rationalization of the scrap-handling sector in Myanmar," Taylor&FrancisOnline, 2020. [Online]. Available: https://www.tandfonline.com/doi/abs/10.1080/01972243.2020.1805248

[15] N. L. M. Azemi, A. A. Kadir, "Design and development of web automated recycle bin system," AIPPublishing, 2023. [Online]. Available: https://pubs.aip.org/aip/acp/article-abstract/2827/1/040001/2910796/Design-and-development-of-web-automated-recycle

[16] M. E. C. Camarillo, L. M. Bellotindos, "A Study of Policy Implementation and Community Participation in the Municipal Solid

Waste Management in the Philippines," AppliedEnvironmentalResearch, 2021. [Online]. Available: https://ph01.tci-thaijo.org/index.php/aer/article/view/240916

[17] H. A. Aziz, S. S. A. Amr, "Strategies for Municipal Solid Waste: Functional Elements, Integrated Management, and Legislative Aspects," SpringerLink, 2022. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-031-06562-0_7

[18] A. A. Mabalay, A. C. Blasa-Cheng, "A Stakeholder Analysis of Manila's Kolek Kilo Kita Waste Management Program," journal.bmu.edu.in, 2024. [Online]. Available: https://journal.bmu.edu.in/journal-files/A_Stakeholder_Analysis.pdf

[19] M. N. Lunag Jr., J. C. Elauria, "Characterization and management status of household biodegradable waste in an upland city of Benguet, Philippines," SpringerLink, 2021. [Online]. Available: https://link.springer.com/article/10.1007/s10163-020-01167-3

[20] M. K. O. Paler, I. D. F. Tabanag, "Elucidating the surface macroplastic load, types and distribution in mangrove areas around Cebu Island, Philippines and its Policy Implications," ScienceDirect, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S0048969722035057

[21] M. Z. Sanchez, P. K. A. Dela Cruz, G. Tagle, R. G. Bautista Jr., R. B. A. Panes, "Clinicord: A Web and Mobile Scheduling System for Medical Clinics in Olongapo City Using Progressive Web App Frameworks," gordoncollege.edu.ph, p 25. [Online]. Available: https://gordoncollege.edu.ph/w3/wp-content/uploads/2024/04/CCS-Research-Journal-2019-2021.pdf#page=30

[22] P. M. Ballaran, C. B. A. Corpuz, "Perazuhan: A Mobile Application for Solid Waste Micro-Management Framework," semanticscholar, 2019. [Online]. Available: https://www.semanticscholar.org/paper/Perazuhan%3A-A-Mobile-Application-for-Solid-Waste-Ballaran-Corpuz/915663ad3a8cc79af1e539dbe28a5eca386cdaeb

[23] W. P. Rey, M. J. G. Fernandez, "CLEANSE: A Web-based Waste Management with a Rewards System," IEEE Xplore, 2024. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10579964

[24] S. N. Domingo, A. J. A. Manejar, "An analysis of regulatory policies on solid waste management in the Philippines: Ways forward," econstor, 2021. [Online]. Available: https://www.econstor.eu/handle/10419/241050

[25] N. Sidhu, A. Pons-Buttazzo, "A Collaborative Application for Assisting the Management of Household Plastic Waste through Smart Bins: A Case Study in the Philippines," MPDI, 2021. [Online]. Available: https://www.mdpi.com/1424-8220/21/13/4534